

Geotechnical and Structures Laboratory (GSL)
Performance Work Statement
Master Blanket Purchase Agreements (BPAs) for Blast Assessments

Branch: Structural Mechanics Branch

Background:

The Structural Mechanics Branch (SMB) of the Geotechnical and Structures Laboratory (GSL) of the U.S. Army Engineer Research and Development Center (ERDC) is conducting research and development (R&D) to support the Department of Homeland Security (DHS). This includes a recurrent need for performing site assessments and blast analyses to provide security recommendations for select sites across the U.S. Multiple BPA Calls for selected sites may be required to be conducted concurrently. **A security clearance level of SECRET and completion of DD-254 is required.** The selected contractor **MUST** be able to provide both the site assessments and the blast analyses. The contractor **MUST** also have access to Vulnerability Assessment and Protection Option (VAPO) software.

Details regarding the tasks for this effort are provided below:

Task 1: Site Assessments and Blast Analyses

Description:

- a) Site Assessments:** Perform site assessment(s) for the selected site(s) to include but not limited to travel to required site, review of existing drawings (as applicable), and capturing all necessary measurements, notes, and photos of building components (e.g., walls, doors, windows, exterior columns, beams, curtain walls) required for blast analyses. These measurements may include glazing thickness of windows/doors, rough openings of windows/doors, and dimensions of window mullions. Data will be collected using a variety of measurement tools (e.g., laser measurer, range finder, tape measure, glass gauge). Floor plans of site(s) may be provided at some site(s) to select critical locations for site assessment (e.g., entrances, pre-function spaces, meeting spaces, locations for VIPs). Prior to the site assessment, a standard operating procedure will be provided.
- b) Blast Analyses:** Perform blast analysis using Vulnerability Assessment and Protection Option (VAPO) software of the selected site based on data collected during site visits. Proficiency is required in using VAPO to develop the comprehensive site map that identifies the required standoffs to meet the security criteria for the selected site. Security criteria will be defined by the Government and

provided to the Contractor. In cases where native analysis modules in VAPO are insufficient for the building element being evaluated, Contractor shall develop a suitable pressure-impulse (PI) curve using other software and import it into VAPO. Other analysis methods include, but are not limited to, Single-Degree-of-Freedom Blast Effects Design Spreadsheets (SBEDS), Wingard, or contractor-developed analysis spreadsheets/software. Analysis will include entire system performance; for example, a window system will require the analysis of the glass, frame and anchorage system/connection points. Additionally, expertise is required in evaluating the response of structural components to blast loads and a complete understanding of blast behavior, including the influence of surrounding structures on blast loads. The blast analysis shall include the influence of surrounding structures on the blast loading environment. Contractor shall also perform a higher fidelity analysis, such as finite element analysis, as required for complex scenarios (e.g., near-field blasts, irregular structural components, complex materials) that cannot be accurately solved using VAPO or other engineering tools.

Deliverables:

a) Site Assessments:

- I. Site assessment notes (Microsoft Word or PowerPoint format) delivered to Government within 2 weeks after the site visit. Notes shall include all measurements, data, and commentary gathered during the site visit. Notes shall include overall photos of the building and site, as well as representative photos of each building element included in the assessment. Notes should be clearly formatted with references so that it is clear to the reader where each building element is located (e.g. key plans, numbering scheme, etc.).
- II. Digital photo log (JPEG format) that includes all photos taken of the site and building components. Photos shall be delivered to Government within 2 weeks after the site visit. Photos shall be transmitted via Contractor file sharing website or via mailed CD.

b) Blast Analyses:

- I. Briefing slides and blast hazard maps (Microsoft PowerPoint format): Contractor shall provide briefing slides for the site that provide an overview of the site, summarize the primary building components that were evaluated in the assessment, and explain any assumptions/limitations of the assessment. Contractor shall also provide blast hazard maps showing the blast threat at each surrounding intersection that will result in an unacceptable failure of the building. Multiple maps may be required showing the threats for various sections of the building, as well as an overall consolidated map showing the minimum threat for the whole building.

Government will provide the applicable security criteria (CUI or SECRET). Contractor shall calculate the threat at up to 50 intersections surrounding the site. Contractor shall work with Government to determine which threat locations will be evaluated for each site. Government will provide a PowerPoint template with applicable formatting requirements. Deliverable may be provided electronically via file sharing site at appropriate classification level or encrypted email.

- II. VAPO analysis files (*.vapoz format): Contractor shall provide exported (*.vapoz format) VAPO models for the analysis that was completed. VAPO models to be provided electronically via file sharing site at appropriate classification level or via mailed CD.
- III. Pressure-impulse (PI) curves (DPlot format): Contractor shall provide PI curves for each type of building element included in the assessment (e.g. window types, façade types, structural elements, etc.). PI curves shall include the corresponding pressure and impulse asymptotes, fit constant, and exponent. PI curves shall be delivered electronically via file sharing site at appropriate classification level or via mailed CD.
 - i. For windows being analyzed, these curves shall provide the lower limits of minimal, low, and high hazard damage as defined by the “UK Glazing Hazard Guide”. See descriptions in table below.

Hazard Level	Description
Minimal Hazard	Low energy glass fragments inside building within 3 feet of window, injuries possible.
Low Hazard	Glass fragments are thrown into building up to 10 feet away, potential serious injuries, fatalities not expected.
High Hazard	Glass fragments are thrown over 10 feet into building at a high velocity, severe injuries with potential fatalities.

- ii. For all non-window components being analyzed, PI curves representing the lower limits of moderate, heavy, hazardous, and blowout as defined in PDC- TR 06-08 Table 2-4. See descriptions in table below.

Table 2-4 Component Damage Levels

Component Damage Level	Description of Component Damage
Blowout	Component is overwhelmed by the blast load causing debris with significant velocities
Hazardous Failure	Component has failed, and debris velocities range from insignificant to very significant
Heavy Damage	Component has not failed, but it has significant permanent deflections causing it to be unrepairable
Moderate Damage	Component has some permanent deflection. It is generally repairable, if necessary, although replacement may be more economical and aesthetic
Superficial Damage	Component has no visible permanent damage

- iii. PI curves shall be provided with pressure (in psi) on the y-axis and impulse (in psi-msec) on the x-axis.
- IV. Threat Summary (Microsoft Excel): Contractor shall submit an Excel file that details the calculated charge weight for each intersection on a per-building system basis (e.g. for each window type, façade type, structural member, etc.). Excel file shall also summarize the controlling threat for each intersection. Deliverable may be provided electronically via file sharing site at appropriate classification level or encrypted email.
- V. Blast assessment summary report (Microsoft Word and PDF formats): This deliverable will not be included for every site and will be determined on a case-by-case basis as defined in the specific BPA Call. Contractor shall write a blast assessment summary report that summarizes all the data gathered for the site, the analysis methods used, assumptions and/or limitations of the analysis, and the results of the blast assessment. Government will provide a template for the report. The full compilation of this report is classified as **SECRET**. Most subsections of the report may be developed unclassified in coordination with the Government and then compiled on a classified computer system. Draft unclassified reports may be provided electronically via file sharing site at appropriate classification level or encrypted email. Final classified reports shall be provided as both paper copies and electronic copies on a CD. Contractor to follow all security requirements in the DD-254 and applicable classification guide(s) for marking and handling of classified materials. The Government retains

unlimited rights to all technical data developed under this contract. This includes all input files and other data used in modeling for each BPA call.

VI. *Blast Analyses deliverables timeline:*

- I. Briefing slides and blast hazard maps: 6 weeks after completion of site visit.
- II. VAPO analysis files, Pressure-impulse (PI) curves, Threat Summary, Draft Blast assessment summary report: 10 weeks after completion of site visit.
- III. Final Blast assessment summary report: 14 weeks after completion of site visit.
- IV. Weekly or bi-weekly meetings will be scheduled between Government and Contractor while Contractor is developing blast analyses to evaluate progress and discuss any assumptions made or questions.
- V. The above timelines may be adjusted in the individual BPA call depending on complexity of the site(s).

PLACE OF PERFORMANCE/PERIOD OF PERFORMANCE

The Master BPA shall be for a five-year ordering period. The place of performance for the Master BPA over the 5- year period will vary and may include the following but not limited to: Washington, DC and surrounding areas, California, and Florida.

The period of performance and place of performance for each BPA Call shall be specified at the BPA Call level.

CONTRACTOR QUALIFICATIONS AND REQUIREMENTS

The Contractor shall provide a Project Manager who shall be responsible for all Contractor work performed under this Performance Work Statement and any resulting BPA Calls. The Project Manager shall be a single point of contact for the Contracting Officer and the Contracting Officer Representative (COR). The name of the Project Manager, and the name(s) of any alternate(s) who shall act for the Contractor in the absence of the Project Manager, shall be provided to the Government as part of the Contractor's proposal. During any absence of the Project Manager, only one alternate shall have full authority to act for the Contractor on all matters relating to work

performed under this contract. The Project Manager, alternates, and any other individuals working on this BPA Call shall have a minimum of **SECRET** level clearance and must complete form DD-254. The Project Manager and all designated alternates shall be able to read, write, speak and understand English. Additionally, the Contractor shall not replace the Project Manager without prior approval from the Contracting Officer. The Project Manager shall have at a minimum a full 4-year professional engineering curriculum leading to a bachelor's degree or the equivalent, and more than 10 years of work experience.

The Contractor shall provide a Principal Engineer who shall oversee all technical work performed under this Performance Work Statement. The Principal Engineer shall have a **SECRET** level clearance. This Principal Engineer shall be a registered Professional Engineer with a Master of Science in engineering, mathematics or science and at least 10 years of experience designing and/or analyzing structures for dynamic and blast loads/effects. Expertise shall include at least five years in design and analysis of facades. The Principal Engineer may also serve as the Project Manager.

****The Contractor must sign a non-disclosure agreement (NDA) provided by the Government prior to beginning Task 1.** All appropriate Contractor personnel are required to sign non-disclosure agreements for sensitive/unclassified and classified information.

The selected contractor **MUST** be able to provide both the site assessments and the blast analyses. The contractor **MUST** have access to Vulnerability Assessment and Protection Option (VAPO) software.

Surveillance Matrix:

Performance Task	Performance Standard and Acceptable Quality Level (AQL)	Monitoring Method	Rating	Incentive/Disincentive for Meeting or Not Meeting the AQL
Task 1: Site Assessment and Blast Analyses	Based upon feedback from the Government Project Leads, Site Assessment activities and Blast Analyses results are deemed comprehensive and technically acceptable. Unless a due date is specified, the deliverable will be completed 60 days after receiving tasking from the Government. Reports and briefings must have minimal typographical or grammatical errors. Minor clarification may be required	Government Project Leads input; COR; Monthly Assessments. Inspection Standard(s): Inspection applies to all standards and submittals and the ERDC makes every effort possible to provide feedback promptly. However, the contractor is advised to allow at least 10 business days for review and return of accepted deliverable(s). Generally, 10	Exceptional, Very Good, Satisfactory, Marginal, Unsatisfactory Rating*	If the Performance Standards are found to be non-compliant the Contractor / Vendor is responsible for correcting the task(s) to a Performance Rating above Unsatisfactory and the resulting action will require completion of the task(s) at no additional cost or extension to the Period of Performance.

		business days is sufficient, however longer review windows may be required for unusually large or complex reviews/responses.		
Financial Reporting/Invoicing	Unless a due date is specified, the invoicing will be due 30 days upon completion of each task. Reports have minimal technical, typographical, or grammatical errors. Minor clarification may be required.	Government Project Leads input; COR; Monthly Assessments	Exceptional, Very Good, Satisfactory, Marginal, Unsatisfactory Rating*	If the Performance Standards are found to be non-compliant the Contractor / Vendor is responsible for correcting the task(s) to a Performance Rating above Unsatisfactory and the resulting action will require completion of the task(s) at no additional cost or extension to the Period of Performance.
Deliverables for Task 1	Based upon feedback from the Government Project Leads, the briefings, maps, analysis files, data files, and summary reports under Task 1 are found to be comprehensive and technically acceptable. Unless a due date is specified, the deliverable will be completed 60 days after receiving tasking from the Government. Reports and briefings must have minimal typographical or grammatical errors. Minor clarification may be required.	Government Project Leads input; COR; Monthly Assessments. Inspection Standard(s): Inspection applies to all standards and submittals and the ERDC makes every effort possible to provide feedback promptly. However, the contractor is advised to allow at least 10 business days for review and return of accepted deliverable(s). Generally, 10 business days is sufficient, however longer review windows may be required for unusually large or complex reviews/responses.	Exceptional, Very Good, Satisfactory, Marginal, Unsatisfactory Rating*	If the Performance Standards are found to be non-compliant the Contractor / Vendor is responsible for correcting the task(s) to a Performance Rating above Unsatisfactory and the resulting action will require completion of the task(s) at no additional cost or extension to the Period of Performance.

Security Requirements:

General security requirements and guidance: The Contractor is responsible for compliance with these security requirements. Questions regarding security matters shall be addressed to the designated Government representative (e.g., Contracting Officer Representative (COR), Requiring Activity (RA) representative, or Contracting Officer (if a COR or other RA representative is not appointed)). Contract personnel are critical to the overall security and safety of US Army Corps of Engineers (USACE) installations, facilities and activities, and security awareness training contributes to those efforts. The Department of Defense (DoD) and Army security training requirements specified below, if applicable, are performance requirements; all applicable contract personnel shall complete initial training within 30 days of contract award or the date new contract personnel begin performance on the contract. Within five business days from the completion of training, the Contractor shall provide written documentation (e.g., email or memorandum) to the Government representative. The documentation shall include the names of contract personnel trained and which training they completed; the Contractor shall maintain training records as part of their contract files and be prepared to provide

copies of training certificates to the Government representative. Contractor personnel and vehicles are subject to search when entering federal installations. Additionally, all contract personnel shall comply with Force Protection Condition (FPCON) measures, Random Antiterrorism Measures (commonly referred to as "RAMs"), and Health Protection Condition (HPCON) measures. The Contractor is responsible for meeting performance requirements during elevated FPCON and/or HPCON levels in accordance with applicable RA plans and procedures --this includes identifying mission essential and non-mission essential personnel. In addition to the changes otherwise authorized by the changes clause of this contract, should the FPCON or HPCON levels at any individual facility or installation change, the Government may implement security changes that affect contract personnel. The Contractor shall ensure all contract personnel are aware of their security responsibilities, including any site-specific requirements identified in local policies or procedures.

REMOVING EMPLOYEES FOR MISCONDUCT OR SECURITY REASONS: The Government may, at its sole discretion (via the Contracting Officer), direct the Contractor to remove any employee from facilities for misconduct or security reasons. Removal does not relieve the Contractor of the responsibility to continue providing the services required under the contract. The Contracting Officer will provide the Contractor with a written explanation to support any request to remove an employee.

Antiterrorism (AT) Level I training: All contract personnel requiring routine access to Army installations, facilities, and controlled access areas, or requiring network access shall complete initial and annual refresher AT Level I awareness training. Online AT Level I awareness training is available at <https://jko.jten.mil/> (website subject to change).

Physical security and access control requirements: All contract personnel requiring physical access to a federal installation or facility shall comply with the access control procedures of that location. Contract personnel requiring unescorted access to meet contract performance requirements on a DoD installation in the US shall be vetted by the installation/facility Provost Marshal/Directorate of Emergency Services/Security Office using the National Crime Information Center-Interstate Identification Index (commonly referred to as "NCIC-III") and Terrorist Screening Database (commonly referred to as "TSDB"). Contract personnel shall comply with all personal identity verification requirements specified in installation/facility policies and procedures.

Contract personnel who do not meet requirements for unescorted access to USACE facilities shall coordinate escorted access with the Government representative, as needed. Contract personnel who receive keys, access cards, or lock combinations that provide access to government-owned property shall comply with key and lock control procedures of the RA.

Contract personnel requiring a common access card (CAC): Contract personnel will be issued a common access card (CAC) only if duties involve one of the following: (1) both physical access to a DoD facility and access to DoD information systems or networks; (2) remote access to a DoD information system or network using DoD-approved remote access procedures; or (3) physical access to multiple DoD facilities or multiple non-DoD federally controlled facilities on behalf of the DoD on a recurring basis for a period of 6 months or more. Before CAC issuance, contract personnel must have, at a minimum, a favorably adjudicated Tier 1 investigation or an equivalent or higher

investigation in accordance with applicable Army regulations and Homeland Security Presidential Directive-12 (HSPD-12). At the discretion of the RA, an initial CAC may be issued based on a favorable review of a fingerprint check and a successfully scheduled Tier 1 investigation with the National Background Investigations Bureau. The RA provides contract personnel with additional information and forms to initiate the CAC issuance process, and/or to initiate background investigations, when required. Contract personnel shall complete these processes within established timelines to avoid delays. Contract personnel will be issued a common access card (CAC) only if duties involve one of the following: (1) both physical access to a DoD facility and access to DoD information systems or networks; (2) remote access to a DoD information system or network using DoD-approved remote access procedures; or (3) physical access to multiple DoD facilities or multiple non-DoD federally controlled facilities on behalf of the DoD on a recurring basis for a period of 6 months or more. Before CAC issuance, contract personnel must have, at a minimum, a favorably adjudicated Tier 1 investigation or an equivalent or higher investigation in accordance with applicable Army regulations and Homeland Security Presidential Directive-12 (HSPD-12). At the discretion of the RA, an initial CAC may be issued based on a favorable review of a fingerprint check and a successfully scheduled Tier 1 investigation with the National Background Investigations Bureau. The RA provides contract personnel with additional information and forms to initiate the CAC issuance process, and/or to initiate background investigations, when required. Contract personnel shall complete these processes within established timelines to avoid delays.

Contractor Identification: When Contractor personnel/subcontractors/consultants perform the services required in this Contract on a Government installation, they are required to possess and wear an identification badge that displays their name and the name of their Company. The Contractor shall ensure that Contractor personnel/subcontractors/consultants identify themselves as contractors when attending meetings, answering Government telephones, providing any type of written correspondence, or working in situations where their actions could be construed as official Government acts.

While performing in a Contractor capacity, Contractor personnel shall refrain from using their retired or reserve component military rank or title in all written or verbal communications.

Suspicious Activity Reporting training (e.g. iWATCH, CorpsWatch, or See Something, Say Something): All contract personnel shall receive initial and annual refresher training from the RA representative on the local suspicious activity reporting program. This locally developed training provides contract personnel with general information on suspicious behavior, and guidance on reporting suspicious activity to the project manager, security representative or law enforcement entity.

Contract personnel requiring access to Government information systems: All contract personnel with access to a government information system (including USACE business systems and CAC-enabled websites) shall comply with applicable DoD and Army regulations, and shall use the organization's UserID-Password Administration and Security System (U-PASS) at commencement of services to request network user

accounts. Contract personnel shall complete DoD Information Assurance Awareness training prior to accessing information systems, and annually thereafter.

Training requirements for the protection of sensitive information: All contract personnel with access to critical information (as identified in the RA's OPSEC Program) shall complete initial and annual refresher OPSEC Level I Awareness training, which is available at the following websites: <https://www.iad.gov/ioss/>, or <http://www.cdse.edu/catalog/operations-security.html> (websites subject to change). All contract personnel with access to Controlled Unclassified Information (CUI) shall complete initial and annual refresher CUI training in accordance with applicable Army policy.

Contracts requiring handling or access to classified information: The prime Contractor shall have a Facility Clearance (FCL) at the appropriate level prior to performance on the contract; the RA will sponsor the prime contract company in obtaining the FCL. All cleared contract personnel shall comply with the FCL requirements, as well as applicable laws and regulations regarding contractor access to national security information. For classified contracts, the RA will generate the DD Form 254, which will be attached to the contract.

Threat Awareness Reporting Program: All contract personnel who maintain an active security clearance shall receive initial and annual refresher training on the Threat Awareness and Reporting Program (commonly referred to as "TARP"), provided by a Counterintelligence Agent. As determined by the servicing Counterintelligence Agent for the RA, contract personnel may complete web-based TARP training.

Pre-screen candidates using E-Verify Program: Contractors shall comply with the requirements set forth in FAR clause 52.222-54 Employment Eligibility Verification and FAR Subpart 22.18 in using the E-Verify Program at (<https://www.e-verify.gov/>) (website subject to change) to meet the contract employment eligibility requirements. Contractors are encouraged to cooperate with Federal and State agencies responsible for enforcing labor requirements to include eligibility for employment under United States immigration laws in accordance with FAR 22.102-1(i). An initial list of verified/eligible candidates shall be provided to the COR no later than three business days after the initial contract award. When contracts are with individuals, the individuals will be required to complete a Form I-9, Employment Eligibility Verification, and submit it to the Contracting Officer to become part of the official contract file.

The Government reserves the right to prohibit individuals who are not U.S. Citizens from performing services or delivering goods under this contract.

Advertisements, Publicizing Awards, and News Releases: All press releases or announcements about agency programs, projects and contract (BPA Call) awards need to be cleared by the DHS Program Office and the ERDC Contracting Officer. Under no circumstances shall the Contractor, or anyone on behalf of the Contractor, refer to the supplies, services, or equipment furnished pursuant to the provisions of this contract in any publicity, news release or commercial advertising without first obtaining explicit written consent to do so from the DHS Program Office and ERDC Contracting Officer.

The Contractor agrees not to refer to awards in commercial advertising in such a manner as to state or imply that the product or service provided is endorsed or preferred by the Federal Government or is considered by the Government to be superior to other products or services.

Protection and Safeguarding of Information and Data: Except as specifically authorized by this contract, or as otherwise approved in writing by the CO, all information and data furnished to the Contractor in the performance of this contract, shall be used only in connection with the work under this contract, and shall be protected by the Contractor from unauthorized use, release, duplication, or disclosure.

The Contractor shall take appropriate measures to assure that its personnel, who have or might reasonably have access to such information and data referred to in the paragraph above, agree to honor the Contractor's commitment and safeguard such information and data.

It is further anticipated that in performance of this contract the Contractor may also have access to, be furnished, use, or generate the following types of data (recorded information):

Data of third parties bearing limited rights or restricted rights notices submitted either to the Government or directly to the Contractor; Other data of third parties which the Government has agreed to handle under protective arrangements.

Data, generated by the Government or the Contractor for third parties, for which the Government intends to control the use and dissemination until delivered to the third parties.

In order to protect the interests of the Government, the owners, and the intended recipients of the data described in this section, the Contractor further agrees, with respect to such data described in this section to:

Use and disclose such data only to the extent necessary to perform the work required under this contract, with particular emphasis on restricting the data to employees having a "need to know"; Preclude disclosure of such data outside Contractor's organization performing work under this contract without written consent of the CO; and

Return or dispose of such data as directed by the CO or COR when such data is no longer needed for contract performance.

Nothing contained in this special contract requirement or elsewhere in this contract shall be construed as altering the definition of "technical data" for the purpose of applying the requirements of any applicable FAR clauses.